

# XRPL Validator — Daily Engineering Report · 2026-03-25

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Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

## Summary

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- 2 commits (xrpl-validator 2)

## Engineering changes

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- **fix: pinned-ledger bulk sync, disable background DB writers** ( `a6ec576` , xrpl-validator)
- Bulk sync now pins all workers to a single ledger\_index for consistent state - Split keypace across 3 HTTP endpoints (1 worker each, throttled) - Disabled background fetcher threads that wrote "validated" ledger data into the pinned-ledger DB, corrupting the state snapshot - Disabled close\_ledger deletes that removed entries expecting re-fetch - Added incremental sync module with sweep resume capability - Added ...
- **perf: SHAMap memory optimization — 42GB to 13GB RSS** ( `423028c` , xrpl-validator)
- Eliminate clone-on-insert: insert/delete now mutate in-place instead of deep-cloning the entire 18.7M-node tree on every call - Hash-only leaf nodes: tree stores pre-computed hashes, not full serialized ledger objects (saves ~6GB of duplicated data) - Hash caching on inner nodes: root\_hash() only recomputes dirty paths instead of all 18.7M+ nodes - Fixed-size [Option; 16] array replaces Vec in ...

## Appendix — commits

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| Time  | Repo           | Commit               | Subject                                                     |
|-------|----------------|----------------------|-------------------------------------------------------------|
| 21:47 | xrpl-validator | <code>a6ec576</code> | fix: pinned-ledger bulk sync, disable background DB writers |
| 22:43 | xrpl-validator | <code>423028c</code> | perf: SHAMap memory optimization — 42GB to 13GB RSS         |